Printable Reference: 2D Vector Differential Steering

Input

Target coordinate vector: $\vec{v} = x\hat{i} + y\hat{j}$

1. Target Magnitude (M)

Calculates the overall required speed limit for the maneuver.

$$M = \sqrt{x^2 + y^2}$$

2. Raw Wheel Bias (L_{raw}, R_{raw})

Determines the proportional speed and direction for each wheel. The y-component drives forward/reverse, and the x-component drives rotation.

$$L_{raw} = y + x$$
$$R_{raw} = y - x$$

3. Maximum Scale Factor (S_{max})

Identifies the highest raw wheel speed to use as a normalization denominator, ensuring no wheel exceeds the requested magnitude.

S_{max}=\max(|L_{raw}|, |R_{raw}|)$

4. Final Scaled Wheel Speeds (L_{final}, R_{final})

Normalizes the raw bias and multiplies by the target magnitude to output the final mapped speeds.

$$L_{final} = M \cdot \frac{L_{raw}}{S_{max}}$$
$$R_{final} = M \cdot \frac{R_{raw}}{S_{max}}$$